

Homework — 2.1.1 Thinking Abstractly

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 2.1.1 lesson.

Part A — This week's topic (~14 marks)

1. A delivery company is building software to plan driver routes around a city. They model the city as a graph.

(a) State what the **nodes** and what the **edge weights** should represent so the model can find the *quickest* route. **[2]** (AO2)

(b) Name **one** real-world detail about the streets that the model can safely **discard**, and justify why it is irrelevant to *this* problem. **[2]** (AO2)

2. An airline trains new pilots in a **flight simulator** rather than in a real aircraft.

(a) Give **two** ways in which the simulator, as an abstraction, differs from reality. **[2]** (AO2)

(b) Explain **one** reason the simulator is still effective for training pilots despite these differences. **[1]** (AO2)

3. A weather app shows the public a simple sun/cloud/rain icon for each day, while the underlying system models air pressure, temperature and humidity across a grid. Explain how **abstraction** is being used to present information to the *user*, and why this is appropriate for them. **[2]** (AO2)

4. A student says: "To design this chess app I'll use abstraction — I'll split it into a move-checker, a board-display and a save-game part." Explain why this statement is **wrong**, and name the skill the student is actually describing. **[2]** (AO1/AO2)

5. A theme park is building an app to model the queue for a ride. List **two** details about a visitor the model should **keep**, and **one** it should **discard**, justifying the discard. Then explain in one sentence why removing irrelevant detail helps the developers. **[3]** (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. (1.1.1) State the purpose of the **Program Counter (PC)** during the fetch–decode–execute cycle. **[2]** (AO1)

7. (1.1.3) State **one** advantage of a **solid-state drive (SSD)** over a **magnetic hard disk drive (HDD)**. [2] (AO1)

8. (1.1.2) State what **pipelining** is and one benefit it gives a processor. [2] (AO1)
